Analysis of Course and Degree Results in a University Programme: Relationships between choices of courses in final year and class of Honours Degree

On progress

Panthakan Boonsuriyatham

University of Glasgow

2636785B@student.gla.ac.uk

STATS5029P Statistics Project and Dissertation 2021-22

July 7, 2022

Presentation Overview

Project Description

University undergraduate students are often interested in knowing which combinations of courses are most likely to lead to the award of the highest class of degree.

This project is to study the course choices as well as the results of undergraduate students who enrol level 4 statistical courses in University of Glasgow. We investigate how can the combination of different courses lead to different results.

Project Clarification

The course results of University of Glasgow are on 22-point scale, usually represented by the secondary band - a character and a number.

SCHEDULE A

Primary Grade	Gloss	Secondary Band*	Grade Point	Primary Verbal Descriptors for Attainment of Intended Learning Outcomes
A	Excellent	A1 A2 A3 A4 A5	22 21 20 19 18	Exemplary range and depth of attainment of intended learning outcomes, secured by discriminating command of a comprehensive range of relevant materials and analyses, and by deployment of considered judgement relating to key issues, concepts and procedures
B	Very Good	B1 B2 B3	17 16 15	Conclusive attainment of virtually all intended learning outcomes, clearly grounded on a close familiarity with a wide range of supporting evidence, constructively utilised to reveal appreciable depth of understanding
C	Good	C1 C2 C3	14 13 12	Clear attainment of most of the intended learning outcomes, some more securely grasped than others, resting on a circumscribed range of evidence and displaying a variable depth of understanding
D	Satisfactory†	D1 D2 D3	11 10 9	Acceptable attainment of intended learning outcomes, displaying a qualified familiarity with a minimally sufficient range of relevant materials, and a grasp of the analytical issues and concepts which is generally reasonable, albeit insecure
E	Weak	E1 E2 E3	8 7 6	Attainment deficient in respect of specific intended learning outcomes, with mixed evidence as to the depth of knowledge and weak deployment of arguments or deficient manipulations
F	Poor	F1 F2 F3	5 4 3	Attainment of intended learning outcomes appreciably deficient in critical respects, lacking secure basis in relevant factual and analytical dimensions
G	Very Poor	G1 G2	2 1	Attainment of intended learning outcomes markedly deficient in respect of nearly all intended learning outcomes, with irrelevant use of materials and incomplete and flawed explanation
H			0	No convincing evidence of attainment of intended learning outcomes, such treatment of the subject as is in evidence being directionless and fragmentary
CR	CREDIT REFUSED			Failure to comply, in the absence of good cause, with the published requirements of the course or programme; and/or a serious breach of regulations

Figure: Schedule A, Guide to the Code of Assessment

Project Clarification

When applying the course credits, we can calculate the aggregate score from them. The course results in level 3 and 4 are used to provide an overall aggregate score which is used to determine the degree classification. This classification is the 5-point scale honours degree system.

17.5 to 22.0	First class honours
14.5 to 17.0	Upper second class honours
11.5 to 14.0	Lower second class honours
8.5 to 11.0	Third class honours
0.0 to 8.0	Fail

The board consider separately when the grade point average falls between two of the ranges defined

Figure: The classification awarded by the board of examiners, Guide to the Code of Assessment

Data Overview

This project has two .xlsx files of student records in the following years, collected by Dr Mitchum Bock:

MScProj2019MB1.xlsx
2015/16
2016/17
2017/18

MScProj2020MB1.xlsx
2015/16
2016/17
2017/18
2018/19

The first file **MScProj2019MB1** is completely contained in the second file. We will use only **MScProj2020MB1** in this project.

Data Overview

Year	Level	Programme	Student.Num	S026	...	S008	Credits.Level4	Credits.Level3	Level3Aggregate	Level4Aggregate	Overall.Aggregate	Degree.Classification
2015/16	4	Erasmus	1000057		...	A2	120		21.8	21.42	21.6	1
								⋮				
2015/16	4	Erasmus	1000300		...	A3	120		21.6	19.75	20.7	1
2016/17	4	Erasmus	1000095	A3	...	A2	120		18.9	19.42	19.2	1
								⋮				
2016/17	4	Erasmus	1000243	B2	...	A1	120		20.3	19.42	19.9	1
2017/18	4	Erasmus	1000008	A1	...	A2	120		19.75	14.17	17	2.1
								⋮				
2017/18	4	Erasmus	1000130		...		20			8.50		
2018/19	4	Erasmus	1000329	B2	...	A4	120	#N/A	20.357	18.5	19.4	1
								⋮				
2018/19	4	Erasmus	1000369	A2	...	A4	120	#N/A	18.428	20.08333333	19.3	1
2015/16	4	Maths & Stats	1000062		...		60	60	18	15.33	16.7	2.2
								⋮				
2018/19	4	Stats & Other	1000358	E2	...		60	60	13.17	10.33333333	11.8	

In the selected file, each row contains

1. Year of graduation (2015/16 to 2018/19)
2. Level of courses in the table (always be 4)
3. Degree programme of students (Erasmus, Math & Stats, Single, Stats & Other)
4. Student number
5. Course results (on secondary band)
6. Credits and aggregate scores of Level 3 and 4
7. Overall aggregate score
8. Degree classification (1, 2:1, 2:2, 3)

Missing Data Procedure

Year	Level	Programme	Student.Num	S026	...	S008	Credits.Level4	Credits.Level3	Level3Aggregate	Level4Aggregate	Overall.Aggregate	Degree.Classification
2015/16	4	Erasmus	1000057		...	A2	120		21.8	21.42	21.6	1
								⋮				
2015/16	4	Erasmus	1000300		...	A3	120		21.6	19.75	20.7	1
2016/17	4	Erasmus	1000095	A3	...	A2	120		18.9	19.42	19.2	1
								⋮				
2016/17	4	Erasmus	1000243	B2	...	A1	120		20.3	19.42	19.9	1
2017/18	4	Erasmus	1000008	A1	...	A2	120		19.75	14.17	17	2.1
								⋮				
2017/18	4	Erasmus	1000130		...		20			8.50		
2018/19	4	Erasmus	1000329	B2	...	A4	120	#N/A	20.357	18.5	19.4	1
								⋮				
2018/19	4	Erasmus	1000369	A2	...	A4	120	#N/A	18.428	20.08333333	19.3	1
2015/16	4	Maths & Stats	1000062		...		60	60	18	15.33	16.7	2.2
								⋮				
2018/19	4	Stats & Other	1000358	E2	...		60	60	13.17	10.33333333	11.8	

There are some variables having missing data or NA. We handle some of them:

1. Course results: the student does not enrol in that course
2. Degree classifications: define cutpoints on overall aggregate scores

To work smoothly, I have changed the course results into two datasets.

S026	S007	S002	S023	S019	S006	S016	SProj.30Cr	...	S008
	A1			A1		A1	A1	...	A2
					⋮				
	A2			A1		A5	A3	...	A3
A3				A5		A4	A5	...	A2
					⋮				
B2				B2		A2	A3	...	A1
A1				F2		C1	B3	...	A2
					⋮				
								...	
B2		A4				A5	A5	...	A4
					⋮				
A2		A3				A4	A3	...	A4
			A3			C3		...	
					⋮				
E2		D2	C3					...	

is transformed into

S026	S007	S002	S023	S019	S006	S016	SProj.30Cr	...	S008
	22			22		22	22	...	21
					⋮				
	21			22		18	20	...	20
20				18		19	18	...	21
					⋮				
16				16		21	20	...	22
22				4		14	15	...	21
					⋮				
								...	
16		19				18	18	...	19
					⋮				
21		20				19	20	...	19
			20			12		...	
					⋮				
7		10	12					...	

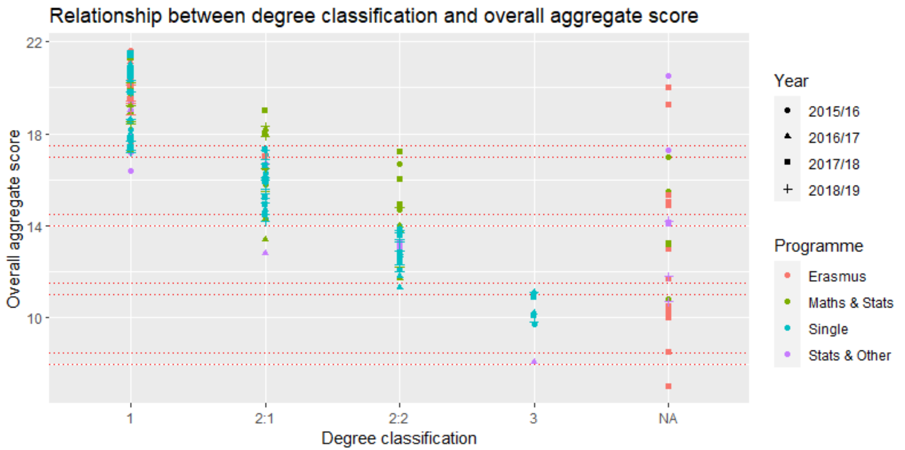
score

and

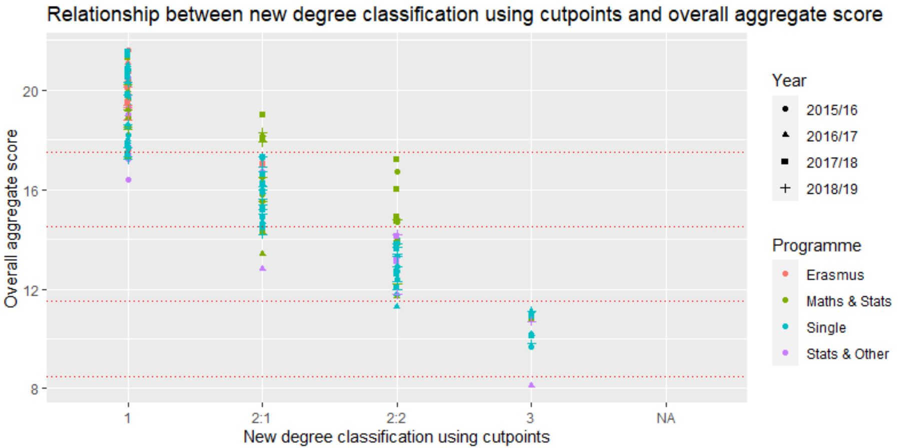
S026	S007	S002	S023	S019	S006	S016	SProj.30Cr	...	S008
F	T	F	F	T	F	T	T	...	T
					⋮				
F	T	F	F	T	F	T	T	...	T
T	F	F	F	T	F	T	T	...	T
					⋮				
T	F	F	F	T	F	T	T	...	T
T	F	F	F	T	F	T	T	...	T
					⋮				
F	F	F	F	F	F	F	F	...	F
T	F	T	F	F	F	T	T	...	T
					⋮				
T	F	T	F	F	F	T	T	...	T
F	F	F	T	F	F	T	F	...	F
					⋮				
T	F	T	T	F	F	F	F	...	F

enrolment status

The overall aggregation score in our data is calculated from the statistics courses each student enrol. Some students who enrol the course outside the school will have a different final overall aggregation score from the data we have. Their degree classification is influenced by other factors we cannot control. Also, some students, especially Erasmus students, do not have degree classifications.



We have created a new degree classification based on the information we have by defining cutpoints. We use the upper bound from Guide to the Code of Assessment to separate, *(17.5, 14.5, 11.5, 8.5)*.



Research Question

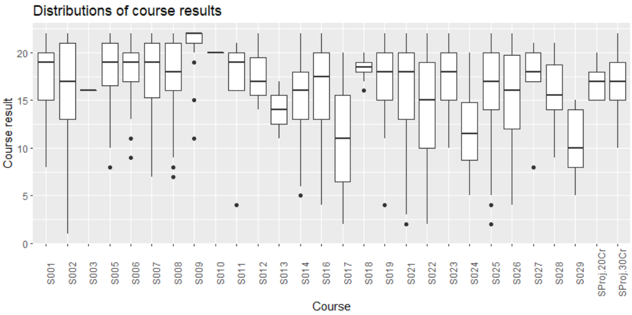
The main research questions of this project are

1. Are there differences in the distribution of course results?
2. Are there differences in the course and degree results between the different programmes of study?
3. Are there any differences in the course and degree results between years of graduation?
4. What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded.
5. What is the relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded? If there is a relationship, which courses should be chosen in the final year of study to maximize the chances of a first class degree?

Question

Are there differences in the distribution of course results?

We answer this problem by using boxplot for the scores of each course.

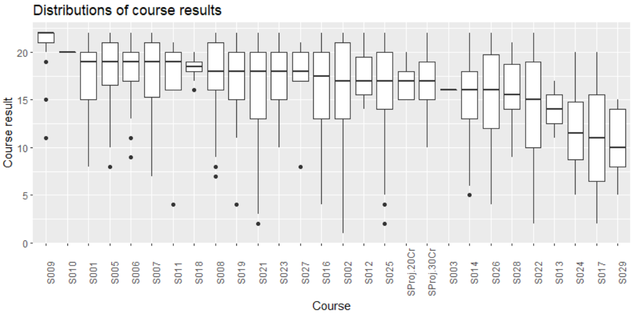


We have 28 courses in total. Some courses contain a lot of data while some contain only a few. We can see differences in median.

Question 1

Question

Are there differences in the distribution of course results?



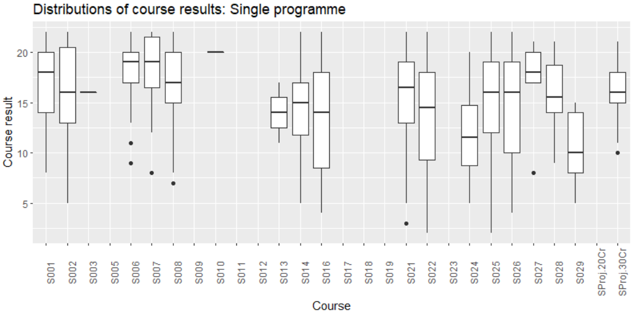
We can see that the median of course results are different. Most of them are left-skewed while few are normal distributed. Surprisingly, the outliers only appear on the left. Over a half of them have students reaching A1.

Question 1

Question

Are there differences in the distribution of course results?

Anyway, the distribution of courses selected in each programme or year are different. The following is an example of single programme students. Some courses have never been selected among the students.

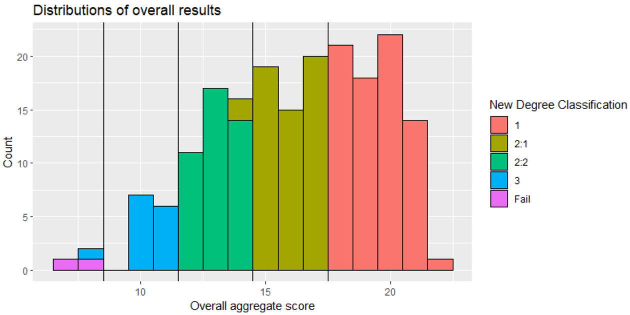


Question 1

Question

Are there differences in the distribution of course results?

It is interesting to have a look at the distribution of overall results as well. We obtain the following figure with the colour of new degree classification.



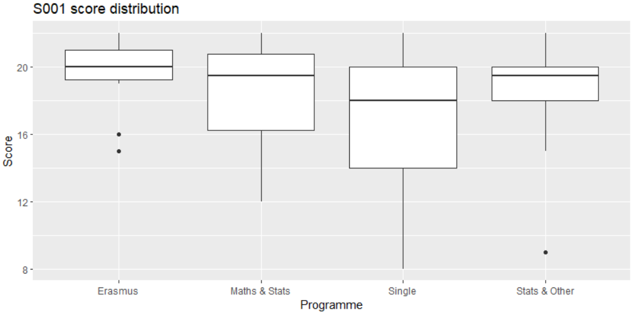
The distribution is left-skewed and show significance of differences.

Question 2

Question

Are there differences in the course and degree results between the different programmes of study?

There are 4 different programmes in our data. For the differences of course, we find that some courses are compulsory while some courses are not chosen in some programme. Only a few number of course that are selected by students in all programmes, for example, S001.

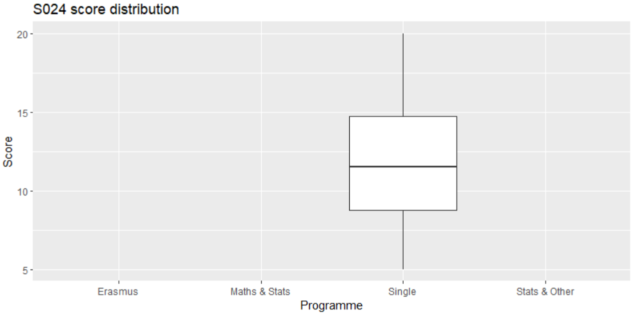


Question 2

Question

Are there differences in the course and degree results between the different programmes of study?

S024 is an example of courses that is only selected by single programme student.

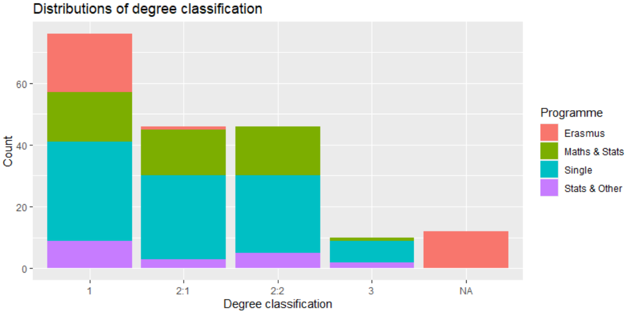


Question 2

Question

Are there differences in the course and degree results between the different programmes of study?

The distribution of degree results by programme can be shown.



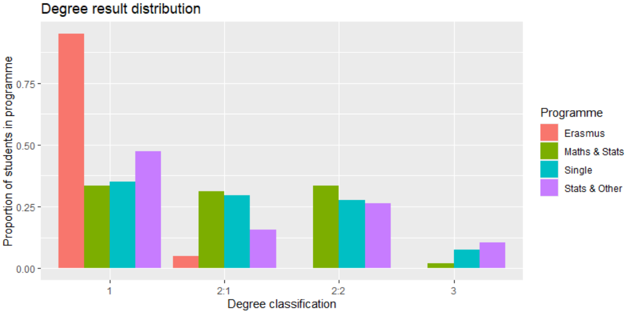
The majority degree classification of erasmus student is first class. The others share the proportion equally.

Question 2

Question

Are there differences in the course and degree results between the different programmes of study?

To compare the proportion, we have the following.



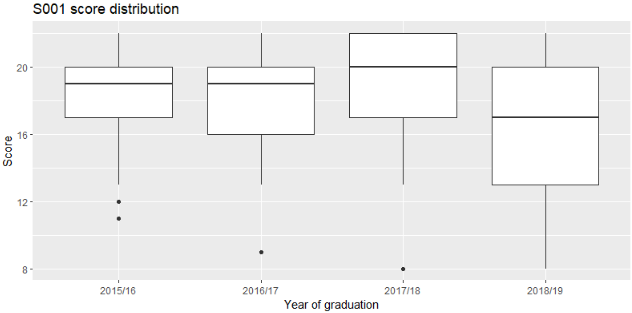
We do not see significant difference between Math & Stats and Single student, but we do for the other pairs.

Question 3

Question

Are there any differences in the course and degree results between years of graduation?

There are 4 different years of graduation in our data. Just like the programme, some courses are selected by students every years, for example, S001.

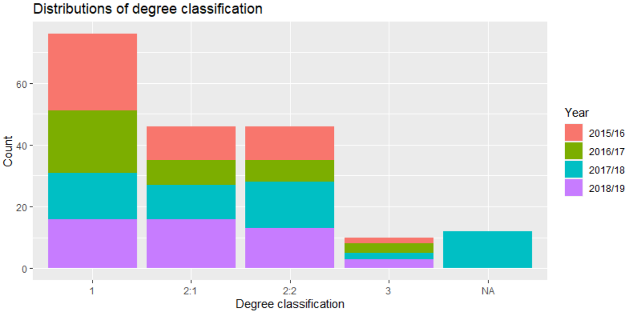


Question 3

Question

Are there any differences in the course and degree results between years of graduation?

The distribution of degree results by year can be shown.



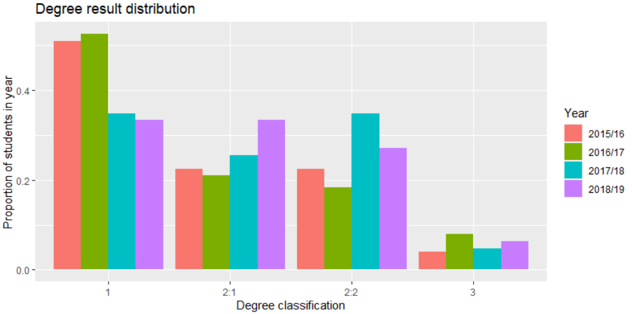
Only in 2016/2017, the number of upper second class students is less than the others. The other years share the proportion equally.

Question 3

Question

Are there any differences in the course and degree results between years of graduation?

To compare the proportion, we have the following.

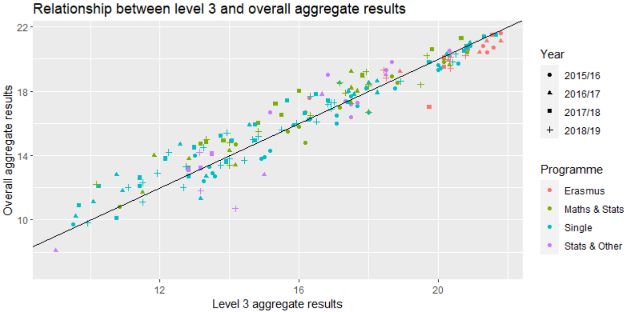


Question 4

Question

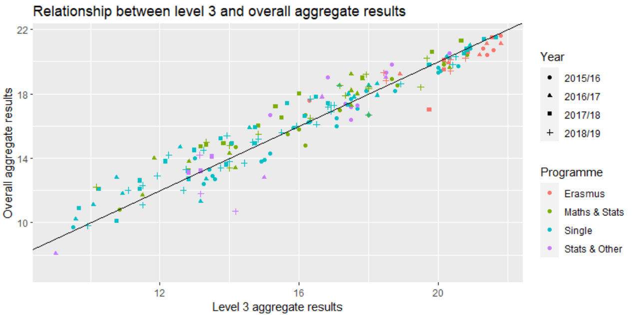
What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded.

To try not to be bias, I omit the missing values from third year instead of assuming the value. The relationship between third year and overall aggregate score are shown below.



Question

What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded.



It is interesting to fit this relationship to a linear model,

$$\text{Overall\_aggregate} \sim \text{Level3\_aggregate} + \text{Programme} + \text{Year}$$

Question 4

The model equation is

$$\text{Overall\_aggregate}_i = \beta_0 + \beta_1 \text{Level3\_aggregate}_i + \text{Programme}_i + \text{Year}_i$$

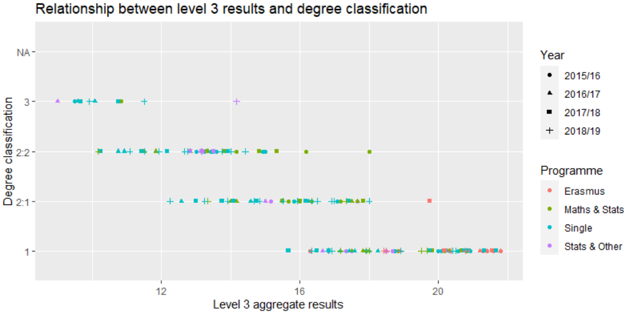
where,

Coefficients	Estimate	Std. Error	P-value
β_0	0.77001	0.53239	0.149926
β_1	0.92756	0.02259	2e-16
Programme: Maths & Stats	0.58444	0.24685	0.019027
Programme: Single	0.24101	0.23853	0.313743
Programme: Stats & Other	0.13397	0.29901	0.654688
Year: 2016/17	0.36972	0.18893	0.051992
Year: 2017/18	0.65223	0.18394	0.000506
Year: 2018/19	0.26068	0.18031	0.150090

Question

What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded.

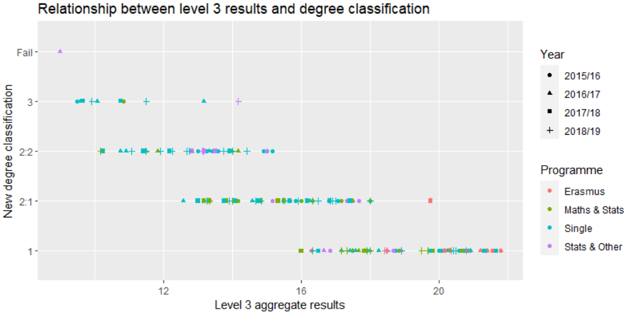
On the other hand, when we are exploring class of degree awarded, we have the original and the new one to consider about. The plot showing relationship of the original classification and level 3 can be shown.



Question

What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded.

Another case is the relationship between the new classification and level 3.



It is also possible to fit ordinal logistic regression.

Question 4

Question

What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded.

We use `polr` funcion from library `MASS` to fit the model. The model is complicated and has 4 equations,

$$\log\left(\frac{p_1}{p_{2:1} + p_{2:2} + p_3 + p_{\text{Fail}}}\right) = \beta_{0,1} - \beta_1 \text{Level3\_aggregate}_i - \text{Programme}_i - \text{Year}_i$$

is an example of them - use to calculate probability of getting first class degree. The accuracy of this model is 141 out of 178, equivalently, 79.21%.

Question 5

Question

What is the relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded? If there is a relationship, which courses should be chosen in the final year of study to maximize the chances of a first class degree?

Obviously, the score of each course has a relationship with aggregate score by a formula. We can use a linear model to answer the choice combination problem. Instead of the overall aggregate score, I decide to use only level 4 instead because of the biasness of level 3, against the enrolment state. There are 2 possible approaches I am considering:

1. A linear model with no course interactions
   $\text{Level4\_aggregate} \sim \text{Course} + \cdots + \text{Course}$
2. A linear model allowing course interactions
   $\text{Level4\_aggregate} \sim (\text{Course} + \cdots + \text{Course}) \times (\text{Course} + \cdots + \text{Course})$

Question 5

Question

What is the relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded? If there is a relationship, which courses should be chosen in the final year of study to maximize the chances of a first class degree?

1. A linear model with no course interactions
$\text{Level4\_aggregate} \sim \text{Course} + \cdots + \text{Course}$

The below table show significant coefficients (95%) in this model

Coefficients	Estimate	Std. Error	P-value
Intercept	13.3111	1.7591	2.7e-12
S009:TRUE	3.0226	0.9354	0.0015
SProj.20Cr:TRUE	2.7979	1.0319	0.0074
S017:TRUE	-6.2272	2.1592	0.0045
S011:TRUE	-3.3192	1.5137	0.0298

Question 5

2. A linear model allowing course interactions
$$\text{Level4\_aggregate} \sim (\text{Course} + \cdots + \text{Course}) \times (\text{Course} + \cdots + \text{Course})$$

The significant variables are less than the first model. The below table show some interesting variables

Coefficients	Estimate	Std. Error	P-value
Intercept	8.5	3.9	0.033
S001:TRUE	7.79	8.7	0.372
S007:TRUE	-32.22	48.4	0.507
S001:TRUE and S007:TRUE	34.1	15.2	0.027
S002:TRUE	63.88	49.6	0.200
S023:TRUE	6.5	4.5	0.149
S002:TRUE and S023:TRUE	-36.2	23.6	0.127
S002:TRUE and S001:TRUE	-48.8	32.0	0.130
S002:TRUE and S022:TRUE	-31.1	20.9	0.139

Question 5

If students expect to have the first class degree classification, it is wise to look for the combinations of first class students. Surprisingly, for non-interactions model, no variables are significant **except for an intercept term**. We find some interesting results from model with interactions.

Coefficients	Estimate	Std. Error	P-value
(Intercept)	17.142	11.5	0.1441
S023:TRUE	9.0	2.9	0.0032
S019:TRUE	4.5	1.8	0.0149
S023:TRUE and S019:TRUE	-6.5	2.9	0.0331
S025:TRUE	-5.3	2.9	0.0733
S026:TRUE and S008:TRUE	-6.0	3.4	0.0830
S026:TRUE and S019:TRUE	-5.3	2.8	0.0621
S002:TRUE and S023:TRUE	-4.0	2.3	0.0872

Model with interactions, first class degree students only

Question 5

Question

What is the relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded? If there is a relationship, which courses should be chosen in the final year of study to maximize the chances of a first class degree?

For the relationship between the choice of final year statistics courses and the class of degree awarded, we can use `polr` funcion from library `MASS` to fit the model.
The new classification is used. The interaction model is not used at this point because the algorithm does not converge.

Question 5

The `summary` function of this model gives coefficients and standard errors of variables. The p-values are not provided so I calculate manually using the formula

$$2 \times (1 - \mathrm{pnorm}(|\frac{\mathrm{coef}}{\mathrm{s.e.}}|, 0, 1))$$

The first few most significant variables are

Coefficients	P-value
S017TRUE	0.00089
S009TRUE	0.00092
SProj.20CrTRUE	0.06717
S011TRUE	0.10935
S018TRUE	0.12611

The model we have does not do well in predicting the results. It predicts correctly 96 out of 190, which is only 51%. We can see that it is because there are not many significant variables.

Question

What is the relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded? If there is a relationship, which courses should be chosen in the final year of study to maximize the chances of a first class degree?

It is worth mentioning a `multinom` funcion from library `nnet` that can be used fit a multinomial model. This model predicts correctly 121 out of 190 (64%). Although it is better than the previous model, it is still not good enough to predict the result and indicate the relationship.

In conclude, there is some relationship. In order to maximize the chances of a first class degree, I suggest taking S009, SProj.20Cr, S001, S007, and S023. However, if students consider enroling additional courses, they should be careful because some combinations can lead to a worse result when adding them together with these course.

The End

Any Suggestions or Comments?